

LLM-Guided Image Recommendation for Supporting Intent Understanding in Non-Verbal Communication

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Abstract

Individuals with intellectual disability and limited verbal abilities often struggle to express complex ideas or initiate conversations. Generic image-based communication boards offer opportunities for open-ended self-expression beyond structured symbols. To support users in constructing and refining visual messages on these boards, this paper presents an image recommendation prototype that provides contextually relevant suggestions to help clarify the intent behind an initial image selection. Powered by large language models (LLMs), the system infers potential communicative intents of the initial image and expands its concepts to support ongoing image-based interaction, rather than relying on linguistic input or symbolic suggestions. The system is evaluated through early-stage quantitative and human assessments of recommendations on a dataset of authentic user-selected images and associated intents. We present early evidence of the system's potential to help users build rich narratives by suggesting diverse images across communicative targets, while noting challenges and opportunities to improve its communicative flexibility.

CCS Concepts

• **Social and professional topics** → **People with disabilities**;
• **Information systems** → **Query reformulation**; • **Human-centered computing** → **Accessibility design and evaluation methods**.

Keywords

Intent-based Image Recommendation, Assistive Communication, Intellectual Disability, Visual Query Expansion, Image-based Communication

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1 Introduction

Generic images, such as photographs, are rich in valuable context that can enrich assistive communication technologies designed for individuals with intellectual disability and minimal verbal abilities. Such images can support self-expression and social interaction [10, 23] without relying on verbal language or symbolic systems like those used in traditional Augmentative and Alternative Communication (AAC) approaches [2]. Prior work has incorporated generic images into interfaces like Visual Scene Displays (VSDs) [1], which offer a more intuitive and flexible means of practicing communication. However, VSDs typically require hotspots pre-defined by speech therapists or family members, limiting users' independence in initiating or navigating conversations around topics of their choosing. The potential of using generic image browsers as open-ended communication aids [10, 11, 23] has also been explored. These systems support communication by enabling users to interactively search for images without using text. The system recommends additional images based on the user's image query, allowing users to arrange them on a board, initiate conversations, and construct messages visually. This provides visual cues that help the audience understand, respond, and ask follow-up questions [10, 11, 23].

Still, a key challenge remains in using image browsers for interactive image search: current image retrieval algorithms focus on retrieving based on visual similarity, not communicative intent, limiting meaningful interaction. Recommending contextually relevant images for a User-selected visual query might support intent clarification and facilitate the audience comprehension [10].

We present an intent-based image recommendation system that leverages LLMs to expand image concepts through the generation of related terms, enabling more relevant image recommendations. This system builds on different aspects of ambiguity in the communicative intent of images—such as ambiguity around locations,



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entities, activities, and occasions—identified in our prior work [11], while also leveraging recent advances in LLMs (e.g., GPT) that have shown potential in interpreting communicative intent and generating conceptual associations [8, 11]. Building on our prior qualitative analysis of how individuals with intellectual disability use generic images for self-expression [10, 11], the recommendation system is meant to be operationalized into a working prototype for non-verbal communication. In the prototype, users first select an image from a screen presenting 20 to 50 images, depending on the configuration. The images are either selected randomly or selected to represent defined categories, from a very large image collection of photographs. The multiple meanings carried by the generic images enable users to always find a visual that relates to what they wish to express, however it can sometimes be related through concept association, or through a small detail of an image rather than its full description. For instance, to find tomatoes, they might select images of pizza, or to find bread, they might refer to images of donuts [10].

The goal of the recommendation system is to allow users to iteratively refine their initial selection until they find an image that best expresses their idea. This open-ended intent expansion represents a significant step beyond topic-specific vocabulary or static intent classification, providing a flexible, user-driven way to express complex ideas visually. Imagine an adult with intellectual disability wanting to express a desire to swim at the beach but unable to verbalize it. With caregiver support, she selects a beach image from a limited image grid, and the system suggests related images like waves, sunshine, or swimmers. By choosing a swimming image, she clarifies her intent, making communication faster and more intuitive than using predefined categories or text.

In this paper, we assess the potential of recommended images through the lens of relevance based on real user data, and with assessors who would be potential communication partners. The more closely the system’s logic aligns with users’ own strategies for accessing and selecting images, the better it can reflect their communicative intents and effectively support meaningful communication. Therefore, the prototype design is informed by data from our earlier co-design workshops with individuals with intellectual disabilities, where 20 user-selected images and their associated communicative intents were collected [11]. It also draws on observations from a case study at a disability organization that examined how generic images are used in communication, highlighting contexts, requirements, opportunities, and challenges [10]. This dataset is expanded through a crowdsourcing project that gathers additional intent guesses beyond the initial user selections. We use this combined data to evaluate how effectively our system can recommend images that help clarify both the actual user intent and alternative intent guesses. Human annotators assess the helpfulness of the recommended images in clarifying intent. Quantitative metrics for intent coverage and diversity are computed to show the system’s potential and identify areas for improvement prior to broader user studies. This early-stage evaluation lays the groundwork for developing more functional assistive technologies that support non-verbal communication.

2 Related Work and Background

2.1 Contextual Image Recommendation for Assistive Communication

Personalized Augmentative and Alternative Communication (AAC) systems and Visual Scene Displays (VSDs) have shown promise in improving user engagement by incorporating familiar environments and customized visual content [1, 17]. Recent work explores contextual visual symbol recommendations for structured activities like dining [21] or board games, and parent-child conversations [4]. Context-aware AAC systems leverage cues to predict and suggest relevant vocabulary [14], though navigating large vocabularies in real-time remains a challenge.

Some studies have proposed generating “just-in-time” communication aids for input photographs, for example, by using visual storytelling dataset as contextual information [7, 13]. Vargas et al. [7] used large language models (LLMs) to provide situation-specific vocabulary and AAC symbols based on visual inputs. However, in these approaches, while users can initiate a conversation with generic images, they must continue it linguistically by selecting appropriate vocabulary and AAC symbols, a process that requires matching abstract symbols to concepts and recalling specific words, which can be challenging for some individuals with minimal verbal ability. Similarly, AccessTalk facilitated parent-child interaction by guiding conversations and suggesting relevant vocabulary cards aligned with the conversational context. It used LLMs to generate parental guidance and curate cards based on the parent’s voice input (transcribed into text) and the child’s input through selected cards [4]. In this app, the conversation still begins with parents selecting topic cards and relies on the child’s ability to understand the symbolic categories (topic, action, and emotion) and match the cards to intended meanings to continue the conversation.

The Canvis web application enabled people with intellectual disability to select generic images through a browsing-based interface as a non-verbal communication method [3, 23]. Canvis was deployed as a technology probe to facilitate communication during National Disability Insurance Scheme (NDIS)¹ planning meetings, helping researchers explore communication control, agency, and engagement in these bureaucratic support processes. Hajizadeh et al. [10, 11] demonstrated the potential of generic image browsers to support personal connection and group participation for people with intellectual disability. Their studies also revealed key challenges in such a prototype, including limited image collections, recommendation algorithms that relied solely on visual similarity, and resulting issues of image accessibility, ambiguity, and intent interpretation. They emphasized the need for image recommendation systems capable of inferring potential user intents from behaviour and feedback, and of providing diverse, meaningful, personalized, and context-aware visual suggestions to enhance communicative effectiveness.

2.2 Intent-based image recommendation

Intent detection has been proposed in image search engines, social media platforms, and conversational agents to support more natural and personalized image recommendations [9, 26, 29]. However,

¹Australian National Disability Insurance Scheme

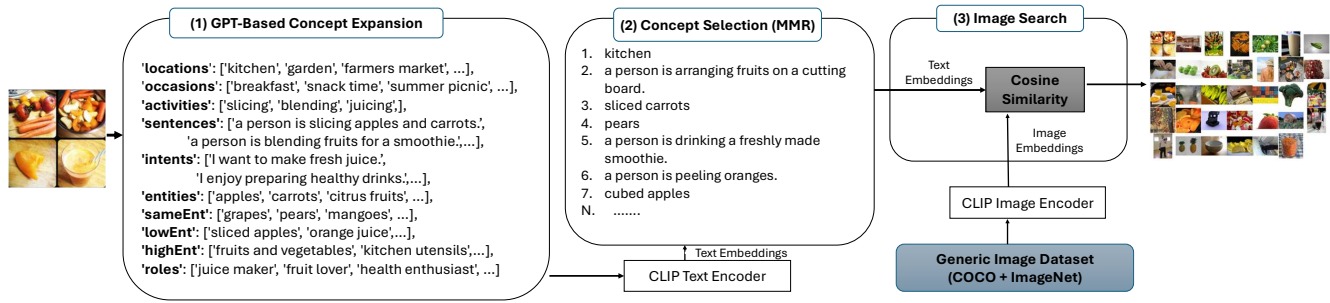


Figure 1: The overall structure of the proposed LLM-Guided image recommendation system for non-verbal communication.

these applications often overlook direct interpersonal communication, especially among users with complex communication needs. Existing studies have used behavioral cues (e.g., clicks, gaze, query logs) for early-stage intent detection during structured tasks [25, 29], but users with complex needs have been ignored in these studies. For marginalized users, intent may vary with context and may be expressed through surface-level visual features rather than abstract queries. In such cases, interactive query suggestion can help clarify actual user intent and support richer conversations [24].

Multimodal approaches that combine visual and textual features significantly reduce the semantic gap in traditional Content-Based Image Retrieval (CBIR) [27, 28]. Models like CLIP (Contrastive Language-Image Pretraining) [22] learn joint embeddings from large-scale image-text data, enabling retrieval systems to align images with diverse, abstract language queries and support more flexible search interactions [6, 16]. Building on this, Ye et al. proposed a user intent expansion framework that jointly parses textual and visual inputs using LLMs and interactive segmentation, composing logic-based search expressions and enabling iterative refinement [30]. Hajizadeh et al. investigated the ability of large language models (LLMs) to interpret image intents and compared these with human perspectives [11]. They found that LLMs, such as GPT-4, closely matched human performance in understanding image intent. Their analysis categorized interpretive guesses by both humans and LLMs into key dimensions, including entity types (e.g., people, objects), visual attributes, activities, temporal context, relationships, locations, and inferred emotions. These approaches mark a shift from similarity-based retrieval to intent-aware systems that better align with users' communicative goals.

3 Methodology

3.1 Recommendation System

We developed an LLM-based image Recommendation algorithm that suggests a new set of images in response to a user-selected image. This image acts as a visual query to recommend related images, helping users clarify or expand their intended message through visually meaningful alternatives. The algorithm comprises three modules: concept expansion, concept selection, and image search (See Figure 1).

In the GPT-based concept expansion module, GPT-4.O interprets the image in terms of communicative intent and expands it into

a diverse set of textual close concepts. The prompt was carefully engineered based on observations from prior co-design studies [10, 11], allowing us to incorporate user perspectives in design and initial assessments.; details are provided in Section 3.2.

In the concept selection module, a simplified version of the Maximal Marginal Relevance (MMR) approach is used to promote diversity and reduce redundancy among generated concepts. Each concept is embedded using CLIP's text encoder, so that its similarity to the initial image, and to other concepts, can be computed. After the concept closest to the initial image is selected, each new concept in the remaining list of proposed expansions is selected so that it is the most dissimilar to those already selected, until a total of 33 concepts are extracted.

In the image search module, the embedding of each selected concept is compared against pre-computed image embeddings from an image database, using cosine similarity for cross-modal retrieval. The image database consists of 5,000 MSCOCO-2017 validation images [19], which feature rich, contextual, multi-object scenes (e.g., events, people, locations), and 50,000 ImageNet validation images, which offer focused object representations. For each concept, the most similar image is retrieved and presented as a visual suggestion. These datasets were chosen because they broadly cover everyday topics relevant for image-based communication.

For each image query, the system recommends 33 new images. This number matches the Canvis prototype [23] used in prior co-design studies. However, it can be adjusted by users according to their preference for viewing fewer or more images at a time.

3.2 Prompt Design and Concept Expansion

The GPT-4 prompt was designed based on prior co-design findings and user data (explained in section 3.3) to reflect the real-world image selection behavior of individuals with intellectual disability (See Appendix B). The prompt integrates three strategies:

(1) **Context Awareness:** We framed GPT-4 as a support worker assisting a person with intellectual disability who communicates using generic images. To mirror the real user scenario, the prompt described the image selection process as relying on a limited set of options and quick choices, rather than assuming a perfect match between image and intended message. It also clarified that the goal of generating concepts was to recommend additional images to

#	Intent guesses in user data
1	I like cooking at home
2	This is my favourite food
3	I love eating food
4	I like mix of food (drinks, meat, vegetables)
5	I want to eat healthy food to lose weight
6	I am hungry
7	I want you to buy all these food
8	I want to eat fruit and vegetables
9	I want you to come and learn how to cut fruits and make milkshakes.
10	I like orange
11	I want you to decorate the food like this and give it to me
12	I want to go for grocery shopping
13	I want to make a juice

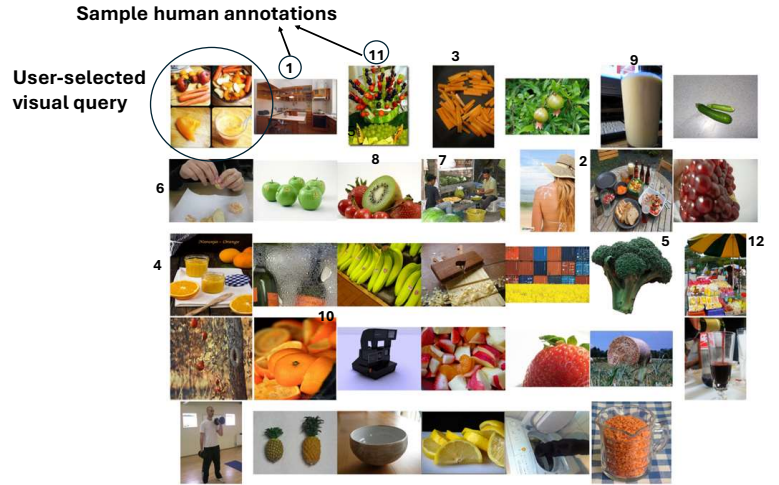


Figure 2: Sample set of images recommended for a user-selected visual query, its intent guesses, and sample human annotations

help clarify the user’s communicative intent. This context helped GPT interpret image intent more naturally and realistically.

(2) **Ambiguity and Conceptual Expansion:** We designed the prompt to capture a range of communicative cues, drawing inspiration from common ambiguities in image-based communication [11] and the associations users are likely to make such as categorical similarity, part-to-whole relationships, and contextual similarities[10]. GPT was asked to generate concepts in eight categories, including relevant locations, entities (and hierarchical variations), activities, occasions, relational roles, intent guesses, and action sentences.

The name of categories and provided examples in the prompt were iteratively refined through testing with user data to ensure alignment with potential communicative intents for user-selected images. For example, "Relational Roles" or "Relevant Occasions" were selected because, for some images, people wanted to talk about their relationships with family, pets, or friends, as well as different occasions such as celebrations or birthdays. The generated intent guesses represent the predicted reasons for selecting an image, while the action sentences represent actions a person might want to take with entities from the image context. For example, if the image includes fruits, the guess can be "I am hungry", and the action sentence is 'a person is slicing fruits and vegetables.'. This can help users to clarify their intent because the main goal for this prompt was to provide the concepts that can be simply visualized by images.

(3) **Diversity and Generalizability:** To avoid bias toward specific themes, the prompt remained general, with instructions to generate diverse, concise (5–10 words per category) concepts. We emphasized concept variety over synonyms to enrich the search space for image recommendation.

3.3 User Data

3.3.1 User-selected visual queries. In prior co-design workshops [11], participants with intellectual disability were invited to select images for self-expression from a simple random image grid

provided by a web-based prototype, named Canvis [23]. The activity was designed as a game: after a participant selected an image, others in the workshop, including friends, support workers, and facilitators, guessed what the image might mean to the participant. The participant could then reveal their actual intent either verbally or by confirming one of the guesses. These interactions formed a dataset of 20 user-selected images, each annotated with a “real” communicative intent and a set of corresponding guesses from the immediate audience. This process defines what we refer to as a user-selected visual query: an image intentionally chosen by a participant to convey a specific, personally meaningful idea or feeling.

3.3.2 Intent guesses. To broaden the dataset and complement the intent guessed during the workshop with guesses from people who did not know the users selecting a queries, additional intents guesses were collected. Twelve postgraduate students were each asked to interpret five of the same set of 20 images, imagining that they had been sent the image by someone with limited verbal ability. Each participant provided at least three possible intent guesses per image, with three responses were collected for each subset. All unique intent guesses from this follow-up were merged with those from the workshop, resulting in a diverse intent set per image—ranging from 7 to 15 intents depending on the number and variation of guesses.

The final dataset thus reflects a mixture of self-expressed, socially interpreted, and crowd-inferred communicative intents. Figure 2 presents one sample image along with its associated intents, while additional examples from the dataset are provided in Appendix A.

3.4 Evaluation

To assess the system’s initial feasibility, we conducted a structured annotation task involving three of the paper’s co-authors who were not involved in the system development. These annotators are experienced in co-designing technologies for individuals with

intellectual disability and are familiar with interpreting communicative intent in visual contexts. In this task, annotators were asked to assess the degree to which each recommended image matched each actual intent as well as each intent guess associated with each user-selected visual query. The evaluation involved two sequential steps:

- (1) Annotators first reviewed the main image, the one originally selected by the user, and rated how well they believed each listed intent genuinely corresponded to that image.
- (2) Then, for each recommended image, they assessed how much it could help clarify the given intent when considered in conjunction with the main image.

In this context, the relevance of the recommended images is defined by their ability to clarify the intended meaning of the main image. That is, an image is considered relevant only insofar as it contributes to disambiguating or reinforcing a particular intent when viewed alongside the original image. Each image–intent pair was rated using a 4-point ordinal scale: 3 – Clearly Clarifies, 2 – Mostly Clarifies, 1 – Somewhat Clarifies, 0 – Unrelated.

The Annotators could evaluate multiple recommended images for each intent, and each image could be associated with more than one intent. This allowed for a more nuanced evaluation of how individual images contribute to different interpretations, acknowledging the potential ambiguity and multiplicity of intent representations.

To examine whether the system was able to suggest images that reflect the diversity of communicative intents users may express, we assess how well the recommended images (1) cover the full range of intents in our dataset and (2) diversify visual interpretations of those intents. These criteria serve as proxies for the system’s potential to support varied user expressions and interpretations, before real-time deployment to identify strengths and limitations in concept expansion and guide further system refinement before larger user-centered studies. We adopted two evaluation metrics adapted from recommender system research [12, 31]:

- **Intent Coverage** – measured as the percentage of intent guesses represented by the recommended images. An intent was considered covered if it had at least one image that received a rating of 2 or 3 by at least two raters.
- **Diversity** – assessed using Shannon Entropy [18], capturing both coverage and distribution across intent guesses.

$$\text{Entropy} = - \sum_{i=1}^n p_i \log_2(p_i)$$

Where p_i is the proportion of assignments to intent i over all assignments across all intents.

4 Results

4.1 Intent-Level Analysis

Before examining the results, it is useful to first consider how valid the different intents in the user dataset were judged to be by our annotators. As described in the evaluation section, one of the tasks was to assess how appropriate the intent guesses were for each image. Based on their feedback, about 62% of the intents were

deemed valid by at least two annotators. Figure 3 presents the valid intents across all user-selected images.

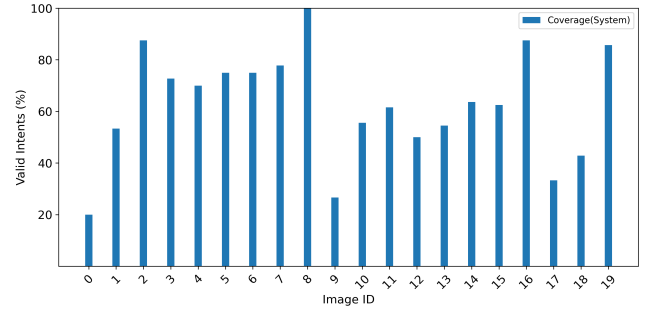


Figure 3: Valid intents per image(b)

A closer examination of invalid intents, especially for images with fewer valid intents (images 0, 1, 9, 17, and 18) revealed key reasons why the annotators marked them as invalid:

- (1) **Abstract or Emotion-Based Intents:** Some intents relied on abstract or emotional interpretations with limited visual cues in the image. For example, the intent *"I am in a dangerous situation and lost in the city"* was linked to an image showing rain in the city, or *"I am tired and I want to rest"* was assigned to an image of a tired-looking cat.
- (2) **Highly Ambiguous Images:** Some images were open to very different interpretations. For example, an image of a toast-like piece with cracked, egg-like shapes was interpreted by the model with geological associations, while human set of intents mostly focused on its edible appearance.
- (3) **The Role of Personal and Cultural Context in Intent Guesses :** Some intent guesses depended on personal or cultural context shared between the image selector and the person guessing the intent – context that could not reasonably be inferred by the system. For example, an image of a pizza selected to imply *"Ninja Turtles love pizza"* relied on background knowledge that the system is not expected to possess.

Therefore, we analyze the results in two scenarios. In the first scenario, metrics are calculated using all intents in the user dataset; in the second, metrics are computed on a refined dataset containing only valid intents. In the main dataset, there is an average of 10.1 intents per image, while in the refined dataset, this value is around 6.25. It is noteworthy that five of the actual user intents provided by users with intellectual disability were also evaluated as "invalid", highlighting a gap between "communicator intent" and "audience interpretation".

4.2 Intent Coverage

Figure 4a shows the distribution of intent coverage percentages for all user-selected visual queries, based on human annotations.

The box plots indicate that intent coverage values for all visual queries ranged from 25% to 100%, with the median of around 65%. This wide range indicates that for some visual queries, the system is

not providing sufficient recommendations that can help the variety of guessed intents. This could be because the system does not understand the possible nuances of the visual query itself, or because it is not able to expand on the concepts gathered from the initial query. Nevertheless, most values exceeded 60%, showing that the system frequently recommended images perceived as meaningful and aligned with intent guesses. For valid intents, most coverage values were above 65%, with a median of around 80%. These results highlight the system’s potential to recommend expressive and relevant visual intents for user-selected queries, particularly when those intents are supported by clear visual cues. When the recommended image set covers most of the plausible intents behind an initial selection, users can refine and clarify their message for a communication partner simply by making additional image choices—reducing reliance on verbal or symbolic explanations.

A closer examination of cases with lower coverage revealed that, in some instances, the system generated a concept aligned with the intent guesses, yet the recommended image did not achieve sufficient agreement among annotators to clearly convey that intent. For instance, the concept *“A person is using a laptop for work”*, generated by the system, aligns closely with the intent *“Looking at laptop too much. His eyes strained, watching YouTube.”* but the recommended image depicted only a laptop without the contextual cues. Similarly, concepts such as *“blanket”* or *“jacket”* were generated for the intent *“I feel cold”* but were not considered sufficiently clear in the context of the main image.

The next question is whether intent coverage is spread across diverse images or concentrated in a few. If the latter, the system may hinder clarity by recommending generic, ambiguous images. We explore this using entropy as a proxy for diversity and balance in concept distribution.

4.3 Entropy

Figure 4b presents the distribution of image entropy derived from human annotations for all user-selected visual queries in both scenarios: including all intents and including only valid intents.

The image entropy, which reflects the diversity of matching intents across recommended images, ranged from 0.41 to 0.94, with a median around 0.7. Removing invalid intents increased entropy values, and some image sets approached the maximum, indicating an even distribution across nearly all available intents. Such balance means that different intents can be expressed and clarified through visually and semantically distinct images, rather than relying on variations of the same concept. By offering meaningfully differentiated suggestions, the system creates opportunities for multiple possible intents to be represented through diverse visuals, supporting clearer message formulation and enhancing the communication partner’s understanding.

4.4 Agreement Analysis

Inter-rater agreement among annotators was assessed using Krippendorff’s alpha, which supports ordinal scales, missing data, and more than two raters. The resulting alpha of 0.53 indicates moderate agreement, falling below the commonly cited threshold of 0.67 for drawing strong conclusions [15]. However, the system’s performance was assessed using a consensus-based criterion where

at least two annotators rated images with 3 (Clearly Clarifies) or 2 (Mostly Clarifies). This approach mitigates the effect of individual variability and emphasizes shared understanding.

5 Discussion

5.1 System’s Interpretation of Communicative Intents for Image Recommendation

We aimed to investigate how effectively our image recommendation system, which leverages LLMs to expand the communicative meaning of image queries, can reflect users’ communicative intents. This investigation seeks to validate the potential for the approach and identify opportunities to improve the quality of the recommendations before conducting user studies. Although the concepts generated by the LLM did not always match the user’s exact intent, they often supported it at the image level, enabling the recommendation of relevant visuals that could help clarify meaning.

Preliminary findings from a small user dataset and annotators indicate that the system shows promising potential to support non-verbal communication by recommending a diverse and intent-rich set of images across various communicative targets. The relatively high intent coverage and entropy, especially for valid intents, suggest that it can help users refine their selections and construct richer narratives. Overall, the results provide early evidence of the system’s feasibility while highlighting its challenges and opportunities to enhance its flexibility and alignment with human interpretation strategies.

5.2 Mismatches between visual queries and actual or guessed intents

This study is an early stage investigation aimed to ensure a good experience for users participating in the user studies that will follow. As a stand alone study however, it presents several limitations regarding the intent list used in the dataset, emphasizing the need for refinement. Only 62% of the annotated intents were approved by our annotators, highlighting inconsistencies in how intents were interpreted. Some intents were overly abstract (e.g., “getting lost”, “dangerous situation”) or dependent on personal or cultural context shared between the image selector and the viewer, context that could not reasonably be inferred by the system. Capturing such personal or culturally grounded meanings may require future work into personal context modeling and user profile integration. Additionally, we found that some Intent guesses resembled image descriptions more than communicative goals. This overlap between descriptive and intentional content complicates the evaluation of whether a second image successfully clarifies the intended meaning of the first.

Furthermore, removing invalid intents left only two or three remaining intents for three of the images—too few for a meaningful calculation of coverage and entropy. In some cases, the user’s actual intent was also excluded for the reasons noted above. This highlights the challenge of how systems might mediate between these two perspectives rather than privileging one over the other. While our aim is to capture real communicative intents and support audience comprehension, within the scope of this study to construct a clear visual message we can only address intents that are precise,

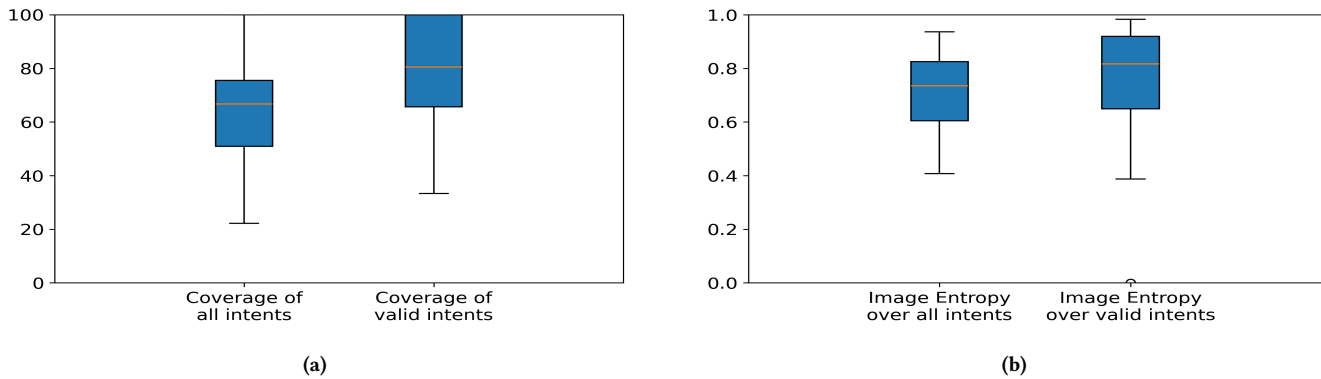


Figure 4: Intent coverage and entropy distributions based on human annotations over all and valid intents.

visually representable, and, for now, not dependent on personal context until such information becomes available to the system. To express more complex intents, users should have the opportunity to clarify their meaning by searching additional images and putting them on the communication boards. For example, if selecting an image of a pizza is intended to start a conversation about "the Ninja Turtles' love of pizza", this meaning cannot be conveyed clearly through image suggestions alone; instead, the user would need to place two different images, one of the Ninja Turtles and one of a pizza, side by side. Similarly, if a single image cannot convey the concept of "holiday", users should have the opportunity to search for and select additional images to communicate the idea more effectively.

Another core challenge lay in the differing mental models and subjective interpretations that humans and the system brought to the same visual content. As discussed in Section 4.1, some images were so visually ambiguous that even human annotators struggled to agree on a clear interpretation. The relatively low inter-rater agreement (Krippendorff's $\alpha = 0.53$) and the perceived invalidity of some actual user intents from the annotators' perspective reflect this variability. While higher agreement would improve reliability, the current approach still provides meaningful insights, particularly given the inherently interpretive nature of the task. Intent-based image recommendation systems must account for the diverse, subjective ways people interpret visuals. Future work should explore deeper personalization and co-design strategies to incorporate user preferences and contextual cues more effectively. This aligns with prior research emphasizing the role of context in augmentative and alternative communication [8, 21].

5.3 Designing for Limitations with Transparency

Some of the aforementioned limitations and challenges cannot be addressed computationally, and should be addressed by improving the design of the interaction, and built into a new version of the prototype before broader user studies. Given the challenges of conveying abstract intents and emotions visually, integrating some textual suggestions generated by GPT alongside images support more effective communication. In turn, these suggestions could

serve as scaffolding, particularly for users who can read, by providing emotional or abstract context that may not be easily inferred from images alone. For instance, if a user selects an image of "a cat wearing a hat" to convey the message "I feel cold", recommending images of a jacket or hat may not sufficiently clarify the intended meaning. In contrast, displaying a textual hint such as "feeling cold" could assist both the image selector and the audience in forming more accurate associations or searching and identifying more relevant images. To support this, we propose separating emotional and intentional concepts from the visual concept selection module and presenting them as text-only suggestions. This enables us to apply MMR to a smaller, more visually grounded concept set, while still enhancing communicative richness through supplemental text.

This proposal also aligns with broader explainability practices in intelligent user interfaces. By explicitly showing the concepts or inferred intents that guided image recommendation, these textual cues can serve as a form of progressive disclosure, helping users and their communication partners understand the rationale behind the system's suggestions. Such transparency is especially critical in assistive contexts, where users and support workers must understand and trust the system's suggestions. Recent work on user reliance calibration and explainable AI interfaces [5, 20] supports this direction, emphasizing the value of making AI-driven decisions more interpretable and context-aware.

Future work will focus on refining the prototype through iterative model development, providing image-text suggestions, and participatory evaluation with individuals with intellectual disabilities.

6 Conclusion

This work introduced a novel intent-based image recommendation prototype to support communication for individuals with intellectual disability. Leveraging large language models to expand image concepts, the system suggests contextually relevant images for a user-selected visual query, aiming to reflect different intents, help clarify it, and support the construction of richer visual narratives on image-based communication boards. Early-stage quantitative evaluation on real user data showed promising results in covering diverse intents, aligning with initial human assessments, and demonstrating potential for future user-centered studies.

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